

FirePumpSim Printable Scenario

Driver/Operator Practice Worksheet

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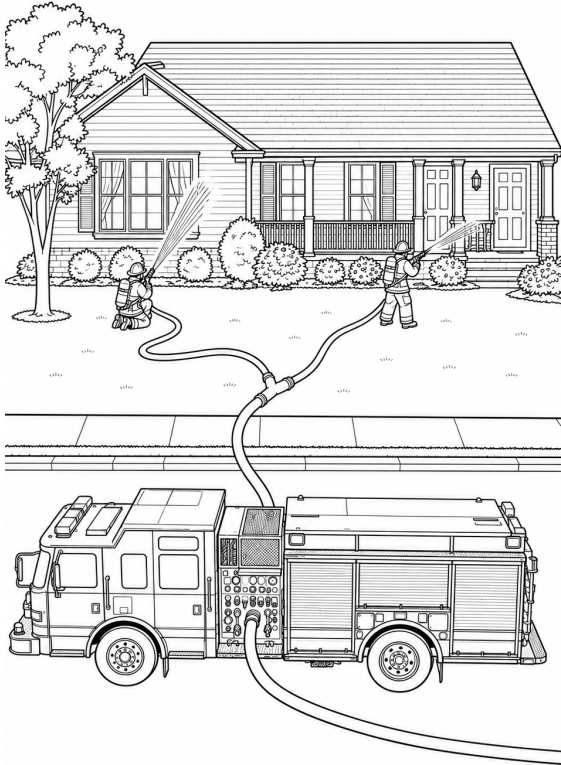
Student Name: _____

Date: _____

Class: _____

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Scenario: Residential Wye Operation - Two Attack Lines



Given / Reference

- Apparatus: Engine 181
- Water source: hydrant supply established
- Main line: 100 ft of 2 1/2 in. hose from engine to wye
- Branch A: 150 ft of 1 3/4 in. hose, 95-GPM fog nozzle at 100 PSI
- Branch B: 150 ft of 1 3/4 in. hose, 95-GPM fog nozzle at 100 PSI
- C values: 1 3/4 in. = 15.5; 2 1/2 in. = 2
- Wye appliance loss: 10 PSI
- Elevation change: 0 PSI

Student Questions

1. What is the total fire flow in GPM when both branch lines are operating?

2. Using $FL = C \times Q^2 \times L$, what is the friction loss in each 150-ft 1 3/4 in. branch line?

3. What is the friction loss in the 100-ft 2 1/2 in. line from the engine to the wye when both branches are flowing?

4. What pump discharge pressure (PDP) is required to supply both branch lines through the wye?

5. If Branch B is shut down and only one 95-GPM branch line is flowing, what new PDP is required?

Required Equations

- $FL = C \times Q^2 \times L$

- $Q = GPM / 100$

- Total Flow = Sum of Operating Lines

- $PDP = NP + \text{Branch FL} + \text{Wye Loss} + \text{Main FL} \pm \text{Elevation}$

- Main Line FL uses total flow through the wye

- Round final PDP to nearest PSI

FirePumpSim Answer Key

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Scenario: Residential Wye Operation - Two Attack Lines



Scenario Givens

- Engine 181 supplies a wye using 100 ft of 2 1/2 in. hose.
- Two branches: each 150 ft of 1 3/4 in. hose with a 95-GPM fog nozzle at 100 PSI.
- C values: 1 3/4 in. = 15.5; 2 1/2 in. = 2.
- Wye appliance loss = 10 PSI. Elevation change = 0 PSI.

Instructor Answer Key

1. Total fire flow with both branch lines

Branch A flow = 95 GPM
Branch B flow = 95 GPM
Total Flow = 95 + 95
Answer: 190 GPM

2. Friction loss in each 150-ft 1 3/4 in. branch line

Formula: $FL = C \times Q^2 \times L$
 $Q = 95 / 100 = 0.95$
 $L = 150 / 100 = 1.5$
 $FL = 15.5 \times (0.95)^2 \times 1.5$
 $FL = 20.99 \text{ PSI}$
Rounded Answer: 21 PSI

3. Main-line FL: 100-ft 2 1/2 in. hose to wye

Total flow through main line = 190 GPM
 $Q = 190 / 100 = 1.9$
 $L = 100 / 100 = 1$
 $FL = 2 \times (1.9)^2 \times 1$
 $FL = 7.22 \text{ PSI}$
Rounded Answer: 7 PSI

4. PDP required to supply both branch lines through the wye

Use the highest branch requirement plus wye loss and main-line FL.
Formula: $PDP = NP + \text{Branch FL} + \text{Wye Loss} + \text{Main FL} + \text{Elevation}$
 $PDP = 100 + 21 + 10 + 7 + 0$
Answer: 138 PSI

5. New PDP if only one 95-GPM branch line is flowing

Branch FL remains 21 PSI. Main flow is now 95 GPM.
Main FL = $2 \times (0.95)^2 \times 1 = 1.8 \text{ PSI}$, round to 2 PSI
 $PDP = 100 + 21 + 10 + 2 + 0$
Answer: 133 PSI

Accept reasonable rounding: +/- 1 PSI.